AdifySkills

n8n Automation Mastery
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CLASS NOTES

🔌 Module 3 · APIs, Webhooks & Integrations

DAY 10 of 25

Databases: Postgres, Supabase, Airtable & Notion

"Sheets se aage — real databases ke saath scale karo."

⚡ AAJ HUM YEH BANAYENGE

Supabase Leads API (Create / Update / Search)

Ek hi webhook se teen kaam: action = create, update ya search. Data Supabase (Postgres) table mein jaata hai aur API JSON lautati hai — apna chhota backend bina server ke.

Postgres · Supabase · Airtable · Notion

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Sheets 10,000 rows ke baad slow ho jaati hain. Real clients ko real database chahiye — Postgres/Supabase.

🎯 Aaj ke baad aap yeh kar paoge

01 SQL basics: SELECT, INSERT, UPDATE, JOIN

🌍 Librarian se: "Delhi wale saare students do" (SELECT), "naya student jodo" (INSERT).
⚙️ SELECT name, phone FROM leads WHERE city = 'Delhi';

02 Postgres/MySQL nodes & Supabase

🌍 Bank ka main computer system.
⚙️ Postgres node → Execute Query / Insert / Upsert; Supabase node → rows CRUD.

03 Airtable & Notion as no-code databases

🌍 Sundar register (Airtable), digital diary (Notion).
⚙️ Airtable → Create record; Notion → Database page create.

04 Upsert patterns & deduplication

🌍 Contact list mein same number do baar save na ho.
⚙️ ON CONFLICT (email) DO UPDATE … / Remove Duplicates node.

05 n8n Data Tables (built-in storage)

🌍 Ghar mein chhota locker — bahar jaane ki zarurat nahi.
⚙️ n8n Data Tables: chhote projects ka built-in storage.

06 Choosing the right DB for a client

🌍 Chhote kaam ke liye cycle, bade ke liye truck.
⚙️ < 5k rows + non-tech → Airtable; app backend / scale → Postgres/Supabase.

⏱ Class ka plan (2 ghante)

10m	35m	45m	20m	10m

`0:00-0:10` **Recap**

🪄 *Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.*

`0:10-0:45` **Concept: Databases**

Sheets vs Airtable vs Postgres

🪄 *Notebook mein 10 naam dhoondhna easy, 10,000 mein mushkil. Database ek super-fast librarian hai.*

`0:45-1:30` **Live Build**

Mini CRM on Supabase with n8n API layer

🪄 *Aaj ki kahani yaad hai — 'Library aur smart librarian'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.*

`1:30-1:50` **Quiz + Challenge**

Module 3 quiz

🪄 *Practice match — jitna yahan girenge, utna asli match mein kam girenge.*

`1:50-2:00` **Q&A + Homework**

🪄 *Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?*

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Table, row, column

Database = Excel jaisa par fast aur reliable.

🪄 **Analogy:** Library ki almirah aur shelves

CONCEPT 02

SQL basics

SELECT * FROM leads WHERE city = 'Delhi'

🪄 **Analogy:** Librarian se sawaal poochna

CONCEPT 03

Upsert

Unique key (email) match → update, warna insert.

🪄 **Analogy:** Attendance: naam hai to tick, nahi to naya naam

CONCEPT 04

No-code DBs

Airtable/Notion — non-tech clients ke liye easy UI.

🪄 **Analogy:** Ready-made furniture

🔑 Aaj ka main idea

Jab data bada ho jaata hai (hazaaron rows), Sheets slow aur risky ho jaati hain. Database (Postgres/Supabase) fast, safe aur multi-user hai. Airtable/Notion beech ka raasta hain — non-tech client ke liye easy UI.

- Sheets = notebook, Database = bank ka ledger system
- Airtable = sundar register jismein filters
- Upsert = attendance: naam hai to tick, nahi to add

📖 Ek kahani se samjho: Library aur smart librarian

1 📚 **Library = database**
= Postgres / Supabase

2 🗂️ **Almirah = table**
= leads table

3 📖 **Kitaab = row**
= Ek lead record

4 🙋 **Librarian se sawaal**
= SQL SELECT

5 🔁 **Kitaab hai to update, nahi to nayi**
= Upsert

🪄 Ek line mein: “Notebook mein 10 naam dhoondhna easy, 10,000 mein mushkil. Database ek super-fast librarian hai.”

🔬 Gehraai se samjho — har concept andar se

Kya hai, n8n andar kya karta hai, kab use karein, kab nahi, example aur common galti.

01 Table, row, column

📖 **Kya hai:** Database ek bahut tez aur reliable Excel jaisa hai. Table = ek sheet (jaise "leads"), column = ek field (name, email), row = ek record (ek lead).

⚙️ **Andar kaise:** Sheet se farak: har column ka type fix hota hai (text, number, date, boolean) — galat type ka data andar hi nahi jaata. Primary key (id) har row ko unique pehchaan deta hai. Unique constraint (jaise email par) duplicate rokta hai. Index se lakhon rows mein bhi search milliseconds mein hoti hai.

✅ **Kab use karein**
Data 10,000+ rows ka ho, kai workflows ek saath likhein, data ko apps/websites se bhi padhna ho, ya relations hon (customer → orders).

⛔ **Kab nahi**
Client ko sirf ek list dekhni hai aur 500 rows hain — Sheet ya n8n Data Table kaafi hai. Database setup bina zaroorat complexity hai.

🧪 **Example:** Supabase table "leads": id (int8, auto), created_at (timestamp), name (text), email (text, unique), status (text, default "new").

⚠️ **Common galti:** Number ko text column mein rakhna (amount = "1200") — fir SUM/ORDER BY galat. Shuru mein hi sahi type chuno.

🪄 **Table = sheet, row = record, column = field (fixed type).**

02 SQL basics

📖 **Kya hai:** SQL database se baat karne ki bhasha hai. 4 shabd 80% kaam kar dete hain: SELECT (padho), INSERT (daalo), UPDATE (badlo), DELETE (hatao).

⚙️ **Andar kaise:** SELECT name, phone FROM leads WHERE city = 'Jaipur' ORDER BY created_at DESC LIMIT 20; — kaunse columns, kaunsi table, kaunsi shart, kis order mein, kitne. Postgres node mein "Execute Query" se SQL likhte ho aur values $1, $2 parameters se dete ho — kabhi bhi user ka text seedha query mein mat jodo.

✅ **Kab use karein**
Custom reports, complex filters, joins (leads + orders), bulk updates. Supabase node ke simple operations kaafi na hon tab.

⛔ **Kab nahi**
Simple create/update ke liye SQL likhne ki zaroorat nahi — Supabase/Postgres node ke ready operations use karo, galti kam hogi.

🧪 **Example:** Postgres → Execute Query: SELECT name, phone FROM leads WHERE city = $1 · Query Parameters: {{ $json.query.city }}

⚠️ **Common galti:** WHERE city = '{{ $json.city }}' likhna — koi user city mein "'; DROP TABLE leads; --" bhej de to poori table ud sakti hai (SQL injection). Hamesha $1 parameters.

🪄 **SELECT padho, INSERT daalo, UPDATE badlo. User input hamesha $1 se.**

03 Upsert

📖 **Kya hai:** Upsert = Update + Insert. Record pehle se hai (jaise same email) to update karo, nahi hai to naya banao. Duplicate kabhi nahi banta.

⚙️ **Andar kaise:** Database ko batana padta hai ki "same" ka matlab kya hai — ek unique column (email, phone, order_id). Postgres mein: INSERT … ON CONFLICT (email) DO UPDATE SET … . n8n mein: Postgres node → Insert or Update; Google Sheets → Append or Update Row (Column to match on); Airtable → Upsert.

✅ **Kab use karein**
Same lead do baar form bhare, webhook retry ho jaaye, roz ka sync — har jagah jahan same data dobara aa sakta hai.

⛔ **Kab nahi**
Jab aapko history chahiye (har payment ki alag row), wahan upsert mat karo — har event ko Insert karo.

🧪 **Example:** Karan ne do baar form bhara → pehli baar INSERT (status new), doosri baar UPDATE (naam theek ho gaya), table mein ek hi row.

⚠️ **Common galti:** Unique constraint lagaye bina ON CONFLICT likhna → error: "there is no unique or exclusion constraint matching the ON CONFLICT specification".

🪄 **Hai to badlo, nahi to banao. Unique column zaroori.**

04 No-code DBs

📖 **Kya hai:** Airtable aur Notion jaise tools database ki taakat ko ek sundar, aasaan screen mein dete hain — non-technical client khud data dekh/edit kar sakta hai.

⚙️ **Andar kaise:** Airtable: base → tables → fields (types: single select, attachment, link to another record). Notion: database pages with properties. Dono ke n8n nodes hain (create, update, search, upsert). Limits: Airtable free plan mein 1,000 records/base aur API rate limit 5 requests/second.

✅ **Kab use karein**
Client ka CRM, content calendar, inventory list — jahan client ko views, filters aur kanban board chahiye.

⛔ **Kab nahi**
Lakhon rows ya bahut tez API calls — Airtable rate limit lagti hai. Wahan Supabase/Postgres + upar se Airtable/Softr interface.

🧪 **Example:** Supabase mein asli data, aur client ke liye Airtable mein ek "Hot leads" view jo n8n har ghante sync karta hai.

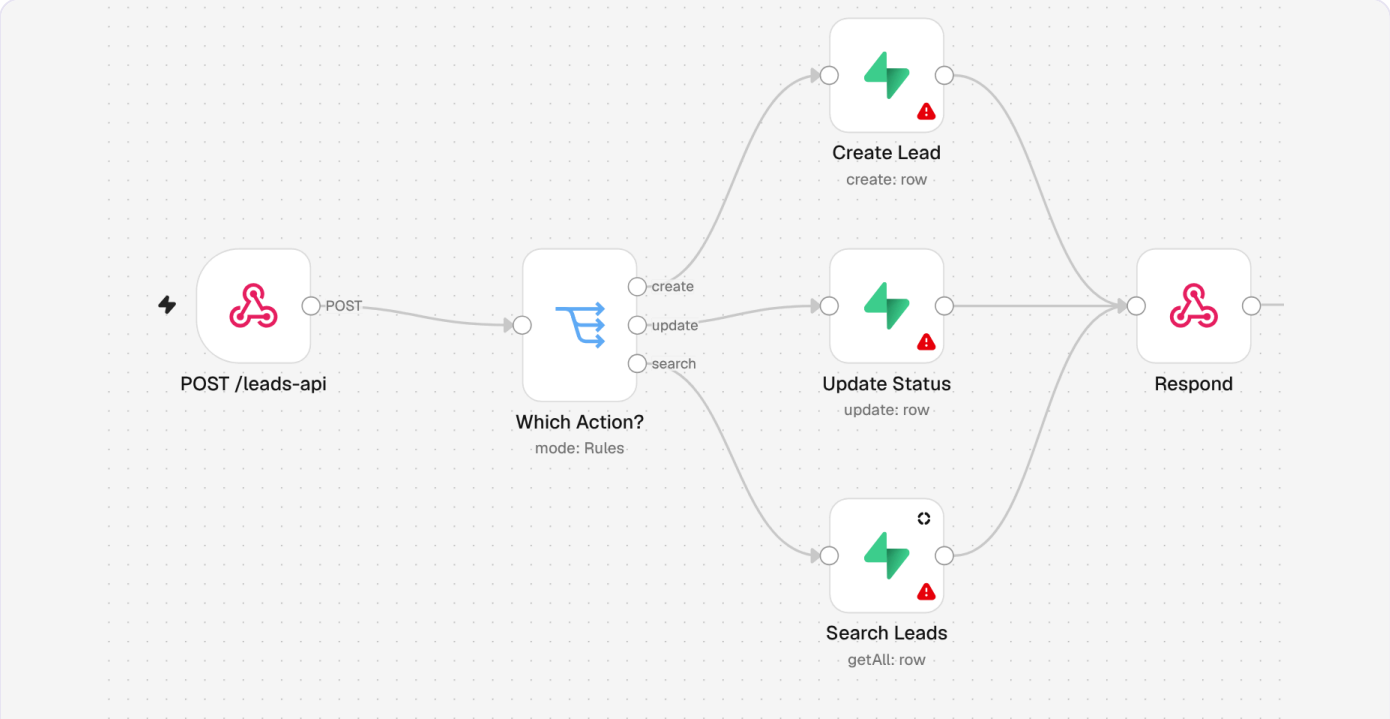
⚠️ **Common galti:** Airtable ka field naam badal diya — n8n mein mapping toot gayi. Live hone ke baad field naam lock rakho.

🪄 **Supabase = engine, Airtable/Notion = sundar dashboard.**

🗺️ Aaj ka workflow: Supabase Leads API (Create / Update / Search)

Data left se right chalta hai. Har dabba ek n8n node hai.





📸 Asli n8n mein import karne par yeh dikhega. Laal ⚠️ = account (credential) connect karna baaki.

💰 Isse kitna kama sakte ho?

Startups (mini backend / leads API) — aam taur par **₹15,000 – ₹40,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Webhook**
POST /crm/lead

2 🗄 **Supabase**
Upsert into leads (email unique)

3 ⚡ **Webhook**
GET /crm/search?city=

4 🗄 **Postgres**
SELECT name, phone FROM leads WHERE city = $1

5 ⚡ **Respond to Webhook**
Results JSON

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- [ ] Connect: **Supabase (project URL + service key)**
- [ ] Supabase SQL editor mein yeh chalao: create table leads (id bigint generated always as identity primary key, name text, email text unique, status text default 'new', created_at timestamptz default now());

✅ Test kaise karein

1. curl se action=create bhejo, phir search, phir update
2. Supabase Table Editor mein data dekho

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Data ka safar — har node ke baad

Aaj ke build mein data kaise badalta hai. Expressions samajhne ki chaabi yahi hai.

1 🌐 POST /leads-api

Body mein action batata hai kya karna hai.

```
{
  "headers": {
    "content-type": "application/json"
  },
  "body": {
    "action": "create",
    "name": "Karan Patel",
    "email": "karan@example.com"
  }
}
```

2 🔧 Which Action?

action = "create" → pehla output. Update/Search branches nahi chali.

```
{
  "create (1 item)": "Karan Patel",
  "update (0)": "—",
  "search (0)": "—"
}
```

3 🔧 Create Lead

Supabase ne nayi row banakar poori row lautayi — id aur created_at database ne khud bhare.

```
{
  "id": 17,
  "created_at": "2026-12-04T09:15:00.512+00:00",
  "name": "Karan Patel",
  "email": "karan@example.com",
  "status": "new"
}
```

4 ↩️ Respond

Caller ko JSON array mila (All Incoming Items).

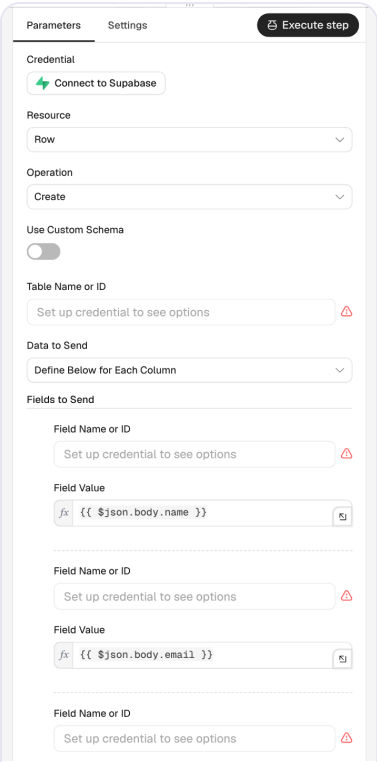
```
[
  {
    "id": 17,
    "created_at": "2026-12-04T09:15:00.512+00:00",
    "name": "Karan Patel",
    "email": "karan@example.com",
    "status": "new"
  }
]
```

Node anatomy — har setting ka matlab

Har zaroori option kya karta hai aur kab badalna hai. Isse aap khud naye workflows bana paoge.

Supabase — Create

Table mein nayi row daalta hai.



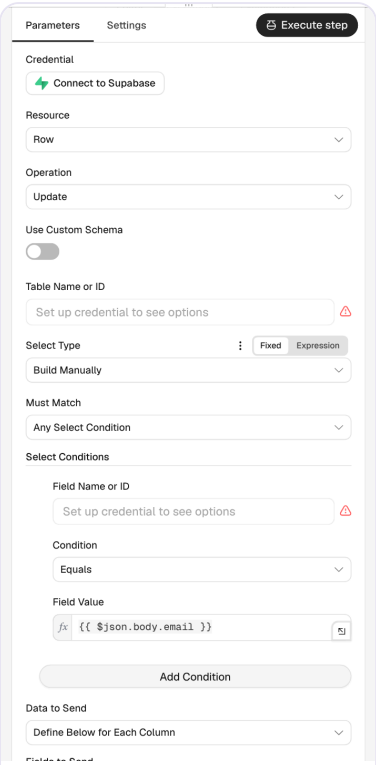
📸 Asli n8n: Supabase — Create

Setting	Kya karti hai	Aaj ki value	Kab badlein
Credential	Supabase project URL + service_role key	`Supabase account`	service_role key sirf server (n8n) par — website par kabhi nahi
Operation	Create, Delete, Get, Get Many, Update	`Create`	—
Table Name or ID	Kaunsi table	`leads`	List se chuno — spelling galti nahi hogi
Data to Send	Define Below (ek-ek field) ya Auto-map	`Define Below`	Auto-map tab jab input fields = columns
Fields	name, email, status	`{{ $json.body.name }} …, status = new`	Default value database mein bhi set kar sakte ho

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Supabase — Update

Email se row dhoondh kar status badalta hai.

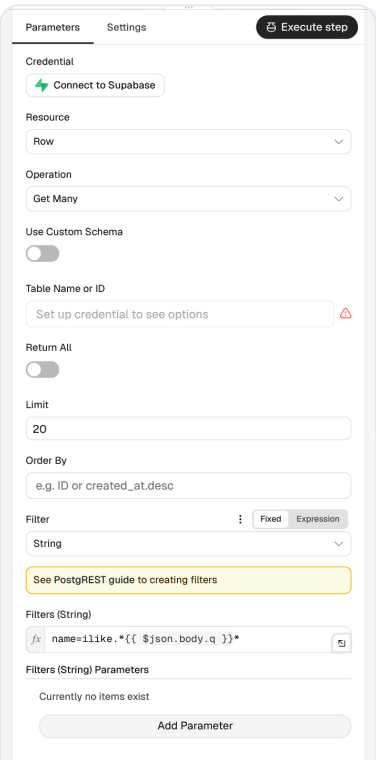


📸 Asli n8n: Supabase — Update

Setting	Kya karti hai	Aaj ki value	Kab badlein
Operation	Update	`Update`	—
Select Conditions	Kaunsi row(s): column + condition + value	`email · Equals · {{ $json.body.email }}`	Shart nahi di to SAARI rows badal sakti hain — dhyan!
Fields to Update	Sirf yeh columns badlenge	`status = {{ $json.body.status }}`	Baaki columns jaise the waise

Supabase — Get Many (search)

Naam ke hisse se leads dhoondhta hai.

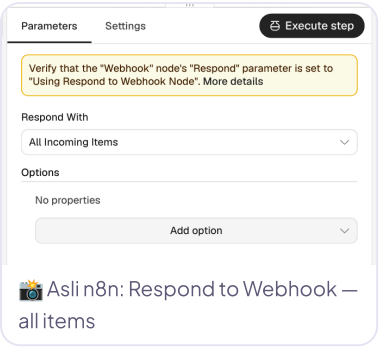


📸 Asli n8n: Supabase — Get Many (search)

Setting	Kya karti hai	Aaj ki value	Kab badlein
Operation	Get Many	`Get Many`	—
Return All / Limit	Kitni rows	`Limit 20`	API mein hamesha limit — warna lakhon rows aa sakti hain
Filter → String	PostgREST filter syntax	`name=ilike.*{{ $json.body.q }}*`	ilike = case-insensitive LIKE; * = kuch bhi
Settings → Always Output Data	Kuch na mile to bhi aage chalo	`On`	Taaki Respond node khaali [] lauta sake

⚡ Respond to Webhook — all items

Jo bhi branch chali, uska result JSON mein lautata hai.



📸 Asli n8n: Respond to Webhook — all items

Setting	Kya karti hai	Aaj ki value	Kab badlein
Respond With	All Incoming Items = saare items JSON array	`All Incoming Items`	Ek hi object chahiye to First Incoming Item
3 branches → 1 Respond	Create/Update/Search teeno isi node se jude	–	Ek execution mein sirf ek branch chalti hai, isliye ek hi jawab
Options → Response Code	200 default	`200`	Create par 201 dena REST ka standard hai

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😫 Pehle (manual)

10,000 rows ki slow Sheet — 10 min

Duplicate dhoondhna — 20 min

30 min

⚡ Ab (n8n)

DB mein fast search + auto dedupe

0 min

⏱ Search seconds mein, duplicates automatic band.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
INSERT INTO leads (email, name, city) VALUES ($1, $2, $3)
ON CONFLICT (email) DO UPDATE SET name = EXCLUDED.name, city = EXCLUDED.city;
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "email": "raj@mail.com",
  "name": "Raj",
  "city": "Jaipur"
}
```

📤 Output — 📋 Select karke copy karo

```
[
  {
    "name": "Raj",
    "phone": "9000011111",
    "city": "Jaipur"
  }
]
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude

📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Ek hi webhook se teen kaam: action = create, update ya search. Data Supabase (Postgres) table mein jaata hai aur API JSON lautati hai — apna chhota backend bina server ke.

Flow: Webhook (POST /leads-api) → Switch (action?) → Supabase (create / update / search) → Respond (JSON result)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt

📋 Select karke copy karo

```
I'm building the n8n workflow "Supabase Leads API (Create / Update / Search)" (nodes: POST /leads-api, Which Action?, Create Lead, Update Status, Search Leads, Respond).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ Workflow JSON
6 nodes · ready to import
`adifyskills.com/workflows/day-10-supabase-leads-api.json`

🌐 Lesson page
Copy buttons, quiz aur video links
`adifyskills.com/day/10`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Supabase (project URL + service key)" connect karo.
Webhook/Form test par kuch nahi ho raha?	"Listen for test event" / Test URL use karo aur 2 minute ke andar request bhejo. Production URL sirf Active workflow par chalta hai.

⚠️ Yeh galtiyan sabse common hain

! SQL mein user input seedha jodna (SQL injection) — query parameters use karo
! Unique constraint ke bina upsert
! Supabase service key ko frontend mein expose karna

⭐ Pro tips — time bachane wali tricks

★ n8n Data Tables chhote projects ke liye built-in storage
★ Index lagao frequently searched columns par
★ Client ke liye Airtable interface bana do — woh khud data dekh sake

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Supabase mein leads table + n8n se create/update/search endpoints.

📝 Homework

Notion content calendar ko Airtable ke saath 2-way sync karo.

☐ Homework ho gaya

BEGINNER
Airtable mein record add ☐

INTERMEDIATE
Notion ↔ Airtable sync ☐

ADVANCED
Paginated search API with LIMIT/OFFSET ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Same email se dobara "create" aaye to error ki jagah "already exists" message lautao.

💡 Hint 1
Pehle search karo, phir decide karo create karna hai ya nahi.

💡 Hint 2
Create se pehle Supabase getAll (filter email=eq.X) + IF "item mila?"

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

Expression practice sheet

Input dekho aur khaali jagah mein expression likho. Jawab aakhri page par hain.

1. Kaunsa action?

```
{"body":{"action":"create"}}
```

Mera jawab: _____

2. Email saaf karke (match ke liye)

```
{"body":{"email":" Karan@Example.com"}}
```

Mera jawab: _____

3. Action valid hai?

```
{"body":{"action":"delete"}}
```

Mera jawab: _____

4. Search filter string banao

```
{"body":{"q":"kar"}}
```

Mera jawab: _____

5. created_at ko padhne layak

```
{"created_at":"2026-12-04T09:15:00+05:30"}
```

Mera jawab: _____

2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Data padhne ka SQL command?

- (A) DROP
- (B) UPDATE
- (C) INSERT
- (D) SELECT

Q2. SQL injection se bachne ke liye?

- (A) Capital letters
- (B) Long query
- (C) Comments
- (D) Query parameters

Q3. Non-tech client ke liye easy DB?

- (A) Postgres CLI
- (B) Airtable
- (C) Redis
- (D) MongoDB shell

Q4. Upsert ka matlab?

- (A) Backup
- (B) Delete+insert
- (C) Update ya insert
- (D) Sirf insert

📖 Aaj ke naye shabd

Webhook
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

JSON
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

CRM
Customers aur leads ka record rakhne ka system.

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 11

Lead Generation & CRM Automation

"Har lead, har baar, 60 seconds ke andar follow-up."

☐ **Day 10 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → D
SELECT

Q2 → D
Query parameters

Q3 → B
Airtable

Q4 → C
Update ya insert

✅ Challenge ka solution

Which Action? (create) → Supabase getAll (email=eq.{{ $json.body.email }}, Always Output Data on) → IF {{ $json.id }} exists → Respond "already exists", warna Create Lead.

🧪 Expression sheet ke jawab

1. `{{ $json.body.action }}` → **create**
2. `{{ $json.body.email.trim().toLowerCase() }}` → **karan@example.com**
3. `{{ ["create", "update", "search"].includes($json.body.action) }}` → **false**
4. `name=ilike.*{{ $json.body.q }}*` → **name=ilike.*kar***
5. `{{ $json.created_at.toDateTime().toFormat("dd LLL, HH:mm") }}` → **04 Dec, 09:15**

📚 Official docs

- Postgres node ↗
- Supabase node ↗
- Notion node ↗
- Airtable node ↗

▶ Videos

- n8n supabase tutorial ↗
- n8n postgres node ↗

Doubt poochho
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